On Relations and Semantic Obstruction in Describing Hyperbolic Curvature in a Dual-Lattice Beltrami-Klein Context

A Canon-Aware Assessment within Lambda / NSAF / TUFT

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Abstract

Read against the entire Lambda / NSAF / TUFT canon, the proposed correspondence between lens-space spectral determinants and the Beltrami-Klein hyperbolic metric remains coherent as a Tier II structural extension, provided it does not claim that a round lens space and hyperbolic space possess the same metric, curvature sign, or topology. The strongest defensible statement holds that the lens-space spectral construction and the Beltrami-Klein metric instantiate the same lift-twist-reattach registry grammar while retaining distinct local geometries and distinct invariant types. This note separates the verified mathematics, the canon-level structural reading, and the remaining Tier III derivations required for a quantitative coefficient-level bridge.

1. Canon-Aware Verdict

The proposal is substantially coherent as a Tier II structural extension, provided it does not assert

$$L(n,1) \cong H^3/\Gamma,$$

or claim that the lens-space determinant and Klein curvature have already been proved numerically identical.

> *The lens-space spectral construction and the Beltrami-Klein metric instantiate the same lift-twist-reattach registry grammar, while retaining distinct local geometries and distinct invariant types.*

Within the canon, "same carrier" need not mean one common Riemannian manifold. It can mean one common registry operation or structural primitive under different operator-dependent readings. The Hopf-Clifford carrier precedes any particular operator; Laplace, Hodge, Dirac, and Beltrami spectra constitute readings of a carrier rather than autonomous foundations.

2. Revision of the Isolated Audit

2.1 The Status of D(n)

Within the canon, D(n) is identified as the lens-space spectral determinant associated with

$$L(n,1) = S^3/Z_n,$$

with the stated values

$$D(1) = 1.203011, \quad D(2) = 4.806545, \quad D(3) = 10.818229,$$

and the approximation

$$D(n) \approx \zeta(3)n^2.$$

The appropriate criticism is therefore not that D(n) is absent from the canon. Rather, any standalone passage should identify the precise determinant combination or normalization inherited from the TUFT construction so that the claim remains independently readable.

2.2 The Meaning of a Common Carrier

The phrase "same carrier" becomes defensible when interpreted as a common registry form rather than a common metric geometry.

- Concrete spectral carrier: the Hopf shell and its cyclic quotient L(n,1).
- Concrete hyperbolic carrier: a bounded convex domain furnished with the Klein metric, or a hyperbolic quotient.
- Common abstract carrier: a bounded or quotiented registry with discrete indexing, group action, coherence boundary, and a non-flat invariant surviving descent.

The round lens space remains locally spherical, while the Klein metric remains hyperbolic. Under standard normalization,

$$K\_L(n,1) = +1, \quad K\_Klein = -1.$$

This difference does not defeat the framework correspondence. It prevents only the stronger claim that the two constructions define literally the same metric geometry.

2.3 The Existing Canonical Foundation of Lift-Twist-Reattach

The canon already establishes the operation through two rigorous realizations:

$$\textit{extend scalars} \;\rightarrow\; \textit{Galois conjugation} \;\rightarrow\; \textit{field norm},$$

$$\textit{lift to a trivializing carrier} \;\rightarrow\; \textit{holonomy} \;\rightarrow\; \textit{projection or descent.}$$

The proposed spectral-hyperbolic comparison therefore need not establish lift-twist-reattach from nothing. It must show that each side occupies the already-defined slots with clearly named objects and maps.

3. Corrected Analysis of the Three Steps

3.1 Lift

Spectral Reading

The map

$$\pi_n : S^3 \rightarrow L(n,1)$$

is a genuine universal-covering construction. The cyclic quotient can be read through the characters

$$\chi_k(g) = \exp(2\pi ik/n).$$

This satisfies the canonical meaning of lift: restoring degrees of freedom hidden or identified by the quotient.

Hyperbolic Reading

The bounded Klein ball already provides a projective model of hyperbolic space. It is therefore technically imprecise to describe the model itself as being lifted to H^3. A canon-compatible formulation should instead state:

> *Lift the bounded Euclidean-coordinate presentation into its homogeneous hyperbolic realization, or, when a genuine quotient occurs, lift H^3/Γ to H^3.*

The Klein metric on the unit ball may be written

$$g_x(u,v) = (u\cdot v)/(1-||x||^2) + ((x\cdot u)(x\cdot v))/(1-||x||^2)^2.$$

The first construction is a representational lift; the second is a topological lift. They should not be silently conflated.

3.2 Twist

Spectral Reading

The genuine twist consists of the Z_n-character or holonomy data:

$$\chi_k \leftrightarrow overline(\chi_k) = \chi_{-k}.$$

This aligns with the canon's conjugate-pair structure and with the left/right Hopf actions represented by the diagonal families (1,1) and (1,-1).

> *Conjugation exchanges complementary phase sectors whose recombination affects the descended spectral invariant.*

Hyperbolic Reading

The twist cannot be identified merely with the negative sign in $1-r^2$. That denominator belongs to the coordinate expression of the metric and does not by itself define an involution or holonomy.

To occupy the canonical twist slot, the dual-lattice model should specify a concrete complementary coupling, for example

$$J:(\theta_1,\theta_2) \mapsto (\theta_1,-\theta_2), \quad J^2 = id,$$

or a left/right phase action on an overlap functional $V\_ov(\theta_1,\theta_2)$. One should then establish a transformation law such as

$$V\_ov \circ J = V\_ov,$$

with oppositely oriented gradient contributions, or a conjugate factorization

$$Q = F\ overline(F).$$

That would provide a genuine twist rather than a suggestive coordinate analogy.

3.3 Reattach

Spectral Reading

For a specified positive elliptic operator A_n, with zero modes removed,

$$log\ det'\ A_n = -\zeta'\_\{A_n\}(0).$$

This plays a norm-like role because it recomposes the regularized mode spectrum into a single descended invariant. Within the canon, it need not be a literal algebraic field norm; it constitutes the spectral realization of the norm/recomposition slot.

Hyperbolic Reading

Projection into the Klein ball recovers a bounded projective representation of the hyperbolic metric. If the overlap functional induces the metric through a Hessian or information-geometric construction,

$$g_{ij} = \partial^2 V\_ov / (\partial\theta^i\, \partial\theta^j),$$

then reattachment can be made mathematically concrete:

$$lifted\ phase\ data \;\rightarrow\; V\_ov \;\rightarrow\; g_{ij} \;\rightarrow\; K(g).$$

Without such a relation, the statement that the norm corresponds to the overlap integral remains Tier III.

4. The Quadratic n^2 Bridge

The values supplied in the canon are

$$1.203011, \quad 4.806545, \quad 10.818229.$$

Dividing by n^2 gives

$$D(1)/1^2 = 1.203011, \quad D(2)/2^2 = 1.20163625, \quad D(3)/3^2 = 1.20202544.$$

Since

$$\zeta(3) \approx 1.20205690,$$

the numerical agreement is strong for the three stated levels. In particular,

$$9\zeta(3) \approx 10.818512,$$

which differs from $D(3)=10.818229$ by approximately 2.6×10^{-5} relatively.

Thus the quadratic law is not merely a vague resemblance inside the paper. What remains unproved in the proposed bridge is the claim that the same index n naturally indexes the hyperbolic geometry and generates the same coefficient.

Hyperbolic ball volume satisfies

$$\mathrm{Vol}\, B(\rho) = \pi(\sinh 2\rho - 2\rho),$$

and therefore, for large ρ,

$$\mathrm{Vol}\, B(\rho) \sim (\pi/2)e^{2\rho}.$$

A quadratic hierarchy follows if the registry derives

$$\rho_n = \log n + c + o(1),$$

because then

$$e^{2\rho_n} \sim e^{2c}\, n^2.$$

This supplies a legitimate bridge mechanism:

$$n \;\rightarrow\; \rho_n \sim \log n \;\rightarrow\; \mathrm{Vol}\, B(\rho_n) \sim n^2.$$

The map $n \rightarrow \rho_n$, however, must arise from the overlap or coherence model rather than being imposed retrospectively. Matching the coefficient $\zeta(3)$ further requires the normalization to emerge from that model.

> *Quadratic spectral residue: supported internally. Hyperbolic quadratic realization: plausible if $\rho_n \sim \log n$. Coefficient-level identity: open.*

5. Semantic Obstruction and the Lambda Residue

The correspondence becomes strongest when expressed through the Lambda principle rather than through equality of curvature signs.

On the spectral side, quotienting and holonomy alter the distribution of admissible modes. The determinant records what survives after regularized recomposition.

On the Klein side, a bounded Euclidean coordinate carrier cannot globally flatten hyperbolic distance. The intrinsic radial distance is

$$\rho(r) = \operatorname{artanh} r,$$

and therefore

$$\rho(r) \rightarrow \infty \quad \text{as } r \rightarrow 1^-.$$

In Lambda language, both constructions display a non-removable departure between the bounded or discretely indexed registry and the global relational structure represented through it.

> *A nonzero global correction survives every locally convenient representation because quotient identification, holonomy, or boundary coherence prevents a globally flat transcription.*

This can be aligned with the canon's residue functional

$$I(x;T) = \inf_{r \in R(x)} O(r,x;T),$$

provided that the target T, admissible representation class R(x), and obstruction functional O are explicitly specified.

The word "semantic" matters here. The obstruction does not reside only in geometry. It also appears in the impossibility of translating all relational content into a bounded, locally flat, or operator-specific description without remainder. The metric and spectral readings need not coincide term by term for them to expose the same failure of complete frame-internal flattening.

6. Correct Tier Placement

Tier I - Verified Components

- $L(n,1)=S^3/Z_n$.
- Character and holonomy decomposition under Z_n.
- Zeta-regularized determinants of elliptic operators.
- Lens-space determinant formulas involving $\zeta(3)$.
- The Klein metric and its constant negative curvature.
- Hyperbolic boundary-distance divergence.
- The exact algebraic and topological prototypes of lift-twist-reattach.

Tier II - Structural Result

The two constructions instantiate the same abstract operation:

restore hidden degrees of freedom → apply conjugate or holonomy coupling → descend to an invariant.

This remains consistent with the canon's 'registries, not privileged maps' posture. Distinct mechanisms need not be declared one object merely because they share a registry grammar.

Tier III - Open Derivation

- Deriving the Klein metric from V_ov.
- Deriving the shell map ρ_n.
- Proving that hyperbolic growth produces the same n^2 term.
- Deriving the coefficient $\zeta(3)$ from the hyperbolic or overlap side.
- Establishing the proposed physical hierarchy beyond the already-audited charged-lepton calculation.

7. Canon-Aware Status of the Spectral-Hyperbolic Correspondence

The lens-space spectral determinant and the Beltrami-Klein metric are not asserted to define the same Riemannian geometry. The round lens space remains locally spherical, while the Klein ball represents constant negative curvature. Their proposed unity lies instead at the level of registry operation.

Each construction admits a lift from a constrained or bounded description into a higher-capacity carrier; each carries a conjugate, holonomy, or complementary phase action; and each descends to a global invariant that cannot be removed by a change of local representation. On the lens-space side, that invariant appears through the zeta-regularized spectral determinant. On the hyperbolic side, it appears through the intrinsic metric and its boundary-divergent distance, or through an overlap functional from which that metric may be derived.

Accordingly, the verified Tier II statement is that both systems instantiate the lift-twist-reattach grammar:

$$\textit{lift} \;\rightarrow\; \textit{conjugate or holonomy twist} \;\rightarrow\; \textit{invariant descent.}$$

The stronger quantitative identification remains Tier III. In particular, the lens-space values used in the TUFT charged-lepton construction closely follow

$$D(n) \approx \zeta(3)n^2,$$

but a hyperbolic realization of the same law requires a derived correspondence between the discrete shell index n and hyperbolic radius or overlap level. A relation of the form

$$\rho_n = \log n + c + o(1)$$

would convert hyperbolic exponential growth into quadratic shell growth,

$$\mathrm{Vol}\, B(\rho_n) \asymp n^2,$$

while matching the coefficient $\zeta(3)$ would require the constant and normalization to emerge from the overlap functional rather than from retrospective fitting.

The correspondence therefore supports the canon's operator-agnostic and residue-centered reading without collapsing distinct geometries into one object. What persists across the two descriptions is not a common curvature sign, but a common obstruction grammar: discrete identification or bounded representation fails to globally flatten the relational structure, and the surviving correction reappears as a spectral or metric invariant.

Conclusion

The bridge should be retained, but its exactness claim should be narrowed to the operational structure rather than extended to an uncomputed coefficient-level equivalence. Within the canon, the relation is neither a loose metaphor nor a completed identity. It is a disciplined Tier II correspondence built on Tier I constructions, with a sharply stated Tier III program: derive the dual-lattice overlap functional, derive the shell-to-radius map, and determine whether the same $\zeta(3)n^2$ residue emerges without fitted normalization.